Project Design Phase-II

Technology Stack (Architecture & Stack)

Student Name: Pranay Patil	**Enrollment No:** EN25EL3010049	**Maximum Marks:** 2 Marks
College / University: Medi-Caps University	**College Email:** en25el3010049@medicaps.ac.in	**Date:** October 2026
Project Title: FORMULA 1 Race Intelligence Dashboard Analysis Report (2010 – 2026)		**Domain:** Data Analytics with Tableau

Technology Stack Selection & Architecture Justification

Selecting an enterprise-grade technology stack guarantees high analytical throughput, interactive responsiveness, and seamless reproducibility across grading and portfolio review platforms.

Component	Technology Selected	Version / Specification	Justification & Purpose
Analytics & BI Tool	Tableau Desktop & Tableau Public	v2024.1+	Industry-standard visual analytics engine providing fast LOD calculations, parameter actions, and responsive dark-mode dashboards.
Data Storage & Staging	Microsoft Excel (.xlsx)	OpenXML Format	Houses relational sheets: circuits, standings, lap_intelligence, qualifying, and race_results (6,464 records).
Calculation Engine	Tableau Calculated Fields & LOD	Native Tableau Syntax	Computes running sums, DNF percentage rates, podium ratios, and reliability indices across filtered eras.
Data Extraction & Validation	Python (pandas / openpyxl)	v3.12+	Pre-processing, schema inspection, data integrity checks, and validation of race record distributions.
Version Control & Portfolio	GitHub Repository	Git LFS / Public	Centralized repository hosting project assets, documentation, dataset, twbx, and demonstration video links.
Demonstration & Media	Screen Recorder / OBS Studio	1080p MP4 / Google Drive	Voiceover screen demonstration walking through the 9 visual sheets and solving the 3 business scenarios.